# INTRODUCTION

## GO! DRIED MANGO 250G

Central Retail in Vietnam has been the country's largest retail chain in recent years, operating a network of 70 hypermarkets. As part of its expansion strategy, a new hyper – **GO! LONG TRUONG** – will open in July 2026 in the eastern area of Ho Chi Minh City.

One of the key FMCG products – **GO! DRIED MANGO** has consistently driven high customer traffic and sales performance across existing stores. Therefore, the Supply Chain Manager has requested a S&OP for this product for the first six months of operation (July–December 2026), with the objective of meeting the company's service level targets while minimizing stockout risk.

To support the analysis, historical data from December 2023 to December 2025 was collected. To ensure demographic comparability and realistic demand estimation, omni-channel sales records from three comparable hypers in the Ho Chi Minh City region were used: GO! AN LAC, GO! BINH DUONG, and GO! DONG NAI.

# DESCRIPTIVE

**BOSS:**

I need to know the **safety stocks of GO! Dried Mango weekly** in HCMC area and the **stocks to meet 90% Service Level**. Moreover, **how many items can be sold in our new hyper** at that time?

**SALE_RECORD**

*The recorded of sale by week, include:*

- sku
- date
- channel
- store
- unit_sold
- stock_avaiable
- promotion_flag
- price_unit
- delivery_days
- is_holiday_week
- is_summer
- is_winter
- sku_age
- lifecycle_stage
- lag_1
- lag_2
- rolling_mean_4
- rolling_std_4

**INV_ANALYTIC**

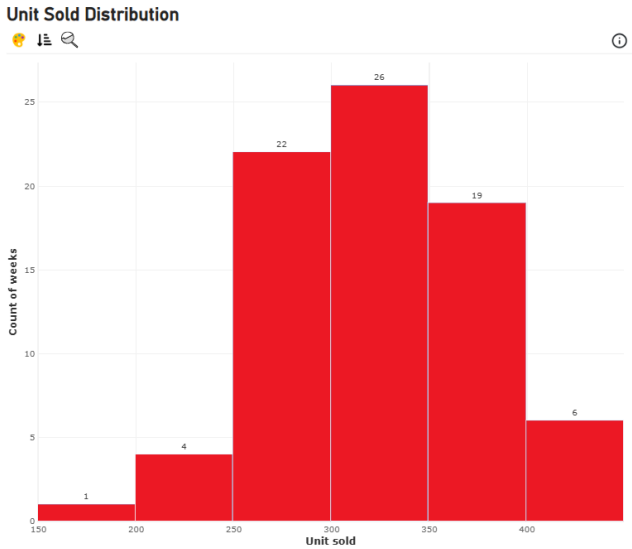
*The recorded of GO! Dried Mango sold from DEC-2023 to DEC-2025 of 3 hypers, include:*

- year
- week_number
- store
- unit_sold

**-> Filtered GO! Dried Mango (RE-025)**

**-> Removed other columns (keep unit_sold, date, store)**

**-> Group by week_number (Sum of unit_sold)**



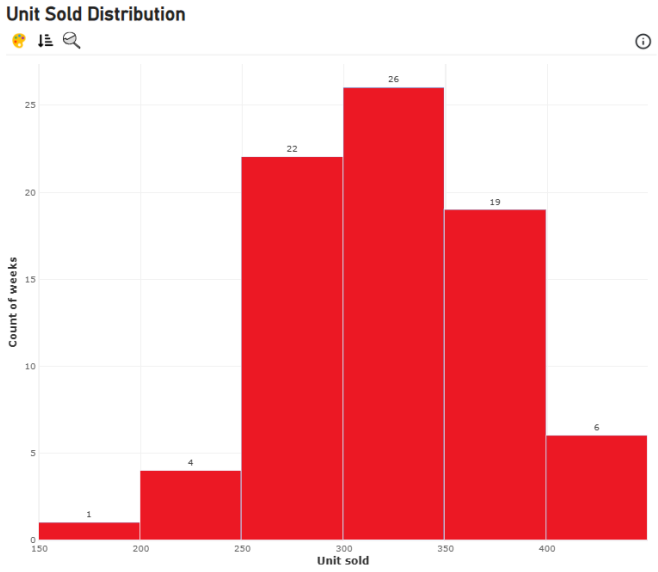

## Data Cleaning

Firstly, I used PCS DW to export sale_recorded.csv file, setting up 3 stores GO! AN LAC, GO! DONG NAI, and GO! BINH DUONG, from DEC-2023 to DEC-2025. It recorded sale information by week and some additional information such as promotion, seasonality, …

## Normal Distribution

| | | | | |
|---|---|---|---|---|
| mean | 325.7051 | | n | 78 |
| std.s | 52.19895 | | max | 419 |
| median | 326 | | min | 176 |
| mode | 325 | | range | 243 |

# DESCRIPTIVE




## 90% Confidence Interval

The average of unit sold is **[316 ; 335]** in HCMC area.

As conclusion, each hyper should have **335** available items on site as safety stock weekly.

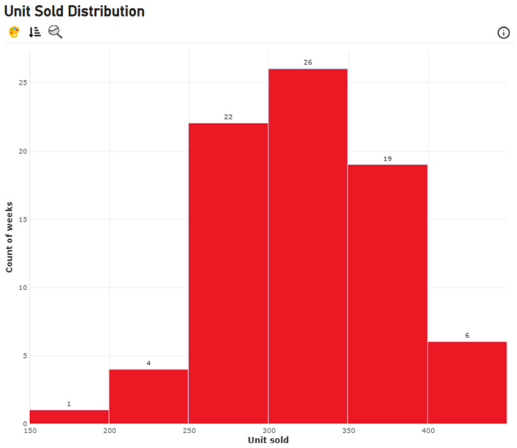
## 90% Prediction Interval

GO! LONG TRUONG can be sold **[240 ; 412]** GO! Dried Mango per week.

## 90% Service Level [CDF(x<X)=90%]

To meet 90% of customer's demand, each hyper should have around **393** available items per week.

# DESCRIPTIVE




| | | | | |
|---|---|---|---|---|
| mean | 325.7051 | | n | 78 |
| std.s | 52.19895 | | max | 419 |
| median | 326 | | min | 176 |
| mode | 325 | | range | 243 |

**BOSS:**

Each hyper in HCMC area can sold **more than 300 GO! Dried Mango items per week as average**, is that true with 5% significance level?

**hypothesis testing**

| | |
|---|---|
| H0 | is not greater than 300 |
| H1 | is greater than 300 |

| | |
|---|---|
| T-test | 4.349159455 |
| p_value | 2.06981E-05 |

**Hypothesis Testing**

Because **p-value < significance** level (0.00002 < 0.05), we **reject Null Hypothesis** (Ho).

So that, **each hyper sold greater than 300 GO! Dried Mango items** weekly with 5% of significance level.

# PREDICTIVE

**BOSS:**

I need to have a **demand planning model** to forecast the demand of GO! Dried Mango.

**SALE_RECORD**

*The recorded of sale by week, include:*

- sku
- date
- channel
- store
- unit_sold
- stock_avaiable
- promotion_flag
- price_unit
- delivery_days
- is_holiday_week
- is_summer
- is_winter
- sku_age
- lifecycle_stage
- lag_1
- lag_2
- rolling_mean_4
- rolling_std_4

**-> Add custom column to binary variables**

**-> Filter store to "GO! DONG NAI"**

**DF_OVERVIEW**

*The recorded of GO! Dried Mango sold from DEC-2023 to DEC-2025 of GO! DONG NAI, include:*

- year
- week_number
- unit_sold
- stock_avaiable
- price_unit
- promotion_flag (binary)
- is mature (binary)
- is decline (binary)
- is_holiday_week (binary)
- is_summer (binary)
- is_winter (binary)
- lag_1
- lag_2
- rolling_mean_4
- roling_std_4
- momentum

→

**SUMMARY OUTPUT**

| *Regression Statistics* | |
|---|---|
| Multiple R | 0.635979763 |
| R Square | 0.404470259 |
| Adjusted R Square | 0.215020099 |
| Standard Error | 31.56833328 |
| Observations | 56 |

ANOVA

| | *df* | *SS* | *MS* | *F* | *ignificance F* |
|---|---|---|---|---|---|
| Regression | 13 | 29104.14864 | 2238.8 | 2.433718524 | 0.014804 |
| Residual | 43 | 42852.06565 | 996.56 | | |
| Total | 56 | 71956.21429 | | | |

| | *Coefficients* | *Standard Error* | *t Stat* | *P-value* | *Lower 95%* | *Upper 95%* | *ower 95.0%* | *pper 95.0%* |
|---|---|---|---|---|---|---|---|---|
| Intercept | -65.75600634 | 64.1143501 | -1.0256 | 0.310812944 | -195.055 | 63.5429 | -195.055 | 63.5429 |
| stock_available | 0.640235455 | 0.23475293 | 2.7273 | 0.009207143 | 0.166811 | 1.11366 | 0.166811 | 1.11366 |
| price_unit | 1.805013331 | 5.350765944 | 0.3373 | 0.737503763 | -8.98583 | 12.59586 | -8.98583 | 12.59586 |
| promotion_binary | 31.8198749 | 9.699809917 | 3.2805 | 0.002059898 | 12.25834 | 51.38141 | 12.25834 | 51.38141 |
| is_mature | -3.206695207 | 21.15131714 | -0.1516 | 0.880205704 | -45.8624 | 39.449 | -45.8624 | 39.449 |
| is_decline | -6.633406014 | 20.22625382 | -0.328 | 0.744533365 | -47.4235 | 34.15672 | -47.4235 | 34.15672 |
| is_holiday_week | 4.672571468 | 13.2176908 | 0.3535 | 0.725434322 | -21.9834 | 31.32859 | -21.9834 | 31.32859 |
| is_summer | -12.41455345 | 13.01605331 | -0.9538 | 0.345522292 | -38.6639 | 13.83482 | -38.6639 | 13.83482 |
| is_winter | 0.772112862 | 17.75042097 | 0.0435 | 0.9655057 | -35.025 | 36.56925 | -35.025 | 36.56925 |
| lag_1 | -0.203646342 | 0.282270403 | -0.7215 | 0.474532357 | -0.7729 | 0.365606 | -0.7729 | 0.365606 |
| lag_2 | 0 | 0 | 65535 | #NUM! | 0 | 0 | 0 | 0 |
| rolling_mean_4 | 0.732062519 | 0.375946424 | 1.9473 | #NUM! | -0.02611 | 1.490231 | -0.02611 | 1.490231 |
| rolling_std_4 | 0.012113756 | 0.482068349 | 0.0251 | 0.980068557 | -0.96007 | 0.984297 | -0.96007 | 0.984297 |
| momentum | 0.298111142 | 0.180255444 | 1.6538 | 0.105446015 | -0.06541 | 0.661631 | -0.06541 | 0.661631 |

→

**Dependent Variable (Y):**

unit_sold

**Independent Variable (Y):**

stock_available
promotion_binary
rolling_mean_4
momentum

**Test - Train:**

Test: DEC-2023 to DEC-2024
Train: JAN-2025 to DEC-2025

## Data Cleaning

Using Power Query to add binary columns for some qualitative criteria. After that, choosing the nearest hyper for GO! LONG TRUONG that GO! DONG NAI. That for fit with customer's behaviors.

## Regression Sample

The main purpose that view the effective of independent variables with unit sold. After that, we can choose some variables have effective with unit sold.

## Model Explanation

As a result, with the recommend of my team, we choose four independent variable with low p-value. We also separated dataset into two parts for train and test.

# PREDICTIVE

**SUMMARY OUTPUT**

| *Regression Statistics* | |
|---|---|
| Multiple R | 0.608297792 |
| R Square | 0.370026204 |
| Adjusted R Square | 0.320616494 |
| Standard Error | 29.81331768 |
| Observations | 56 |

ANOVA

| | *df* | *SS* | *MS* | *F* | *Significance F* |
|---|---|---|---|---|---|
| Regression | 4 | 26625.68482 | 6656.421205 | 7.488937047 | 7.96606E-05 |
| Residual | 51 | 45330.52946 | 888.8339111 | | |
| Total | 55 | 71956.21429 | | | |

| | *Coefficients* | *Standard Error* | *t Stat* | *P-value* | *Lower 95%* | *Upper 95%* | *Lower 95.0%* | *Upper 95.0%* |
|---|---|---|---|---|---|---|---|---|
| Intercept | -68.3031925 | 44.6313953 | -1.5303844 | 0.1321013 | -157.9044574 | 21.2980724 | -157.9044574 | 21.2980724 |
| stock_available | 0.6180711 | 0.2050741 | 3.0138912 | 0.0040106 | 0.2063676 | 1.0297746 | 0.2063676 | 1.0297746 |
| promotion_binary | 28.9025581 | 8.0605897 | 3.5856630 | 0.0007520 | 12.7202491 | 45.0848672 | 12.7202491 | 45.0848672 |
| rolling_mean_4 | 0.6266197 | 0.1731173 | 3.6196252 | 0.0006777 | 0.2790722 | 0.9741671 | 0.2790722 | 0.9741671 |
| momentum | 0.1996080 | 0.0950530 | 2.0999656 | 0.0406979 | 0.0087812 | 0.3904348 | 0.0087812 | 0.3904348 |

| year | week_number | units_sold | test_unit_sold | MAPE |
|---|---|---|---|---|
| 2025 | 1 | 115 | 115.41 | 0.36% |
| 2025 | 2 | 85 | 104.14 | 22.52% |
| 2025 | 3 | 195 | 127.54 | 34.59% |
| 2025 | 4 | 108 | 143.65 | 33.01% |
| 2025 | 5 | 113 | 88.72 | 21.48% |
| 2025 | 6 | 153 | 150.72 | 1.49% |
| 2025 | 7 | 119 | 121.12 | 1.78% |
| 2025 | 8 | 137 | 128.80 | 5.98% |
| 2025 | 9 | 81 | 101.37 | 25.15% |
| … | … | … | … | … |
| | | | | **17.02%** |

## Regression Train

All independent variables have low p-value (< 0.05), so these variables are reliable. The OLS model above explain **37%** the unit sold weekly.

## MAPE Test

Testing the forecast unit sold then check MAPE each week, after that, calculate the average that 17.02%. As conclusion. This OLS Model explain the difference between forecast and reality is **17.02%**.

# PRESCRIPTIVE

**BOSS:**

We spent more than **VND 30 million monthly** for direct logistics from suppliers to hypers last year. We will rent a part of 3PL's warehouse in Thu Duc City, so we have a new plan that non-direct logistics (supplier -> WH -> hyper). You should find the **optimization logistics plan to minimum costs**.

| Sum of unit_sold | Suppliers | | | | |
|---|---|---|---|---|---|
| Hypers | JU | MI | RE | SN | YO |
| GO-AN-LAC | 1304 | 8756 | 7469 | 8718 | 17920 |
| GO-BINH-DUONG | 1249 | 8737 | 7464 | 8607 | 17739 |
| GO-DONG-NAI | 1211 | 8873 | 7458 | 8629 | 17678 |

**LOGISTICS COST** *in carton* *VND 1.000*

| NON-DIRECT | | | | | |
|---|---|---|---|---|---|
| **Suppliers to WH** | JU | MI | RE | SN | YO |
| South WH | 30 | 38 | 41 | 35 | 32 |

| **WH to Hypers** | South WH |
|---|---|
| GO-AN-LAC | 10 |
| GO-BINH-DUONG | 12 |
| GO-DONG-NAI | 13 |

| *about WH* | | |
|---|---|---|
| additional costs | 8000 | |
| capacity | 5000 | cartons |

| DIRECT | | | | | |
|---|---|---|---|---|---|
| **Suppliers to Hypers** | JU | MI | RE | SN | YO |
| GO-AN-LAC | 50 | 70 | 72 | 65 | 60 |
| GO-BINH-DUONG | 60 | 78 | 84 | 77 | 70 |
| GO-DONG-NAI | 58 | 72 | 80 | 70 | 66 |

## 2025 Demand

We have used the Sales Record to combine this pivot table above. To be specific, our Private Labels were supplied by 5 suppliers in the south of Vietnam with 5 different labels but same size.

## Logistics Costs

According to historical shipments, there are the table of logistics costs. In addition, if warehouse open, we must pay additional cost for 5000 cartons capacity.

Furthermore, 1 shipment = 10 cartons = 300 items.

# PRESCRIPTIVE

**VARIABLES'S LOGISTICS COST** *in shipment*

| NON-DIRECT | | | | | |
|---|---|---|---|---|---|
| **Suppliers to WH** | JU | MI | RE | SN | YO |
| South WH | | | | | |

| **WH to Hypers** | South WH | | | | binary |
|---|---|---|---|---|---|
| GO-AN-LAC | | | *about WH* | open/not | |
| GO-BINH-DUONG | | | additional costs | 8000 | |
| GO-DONG-NAI | | | capacity | 5000 | cartons |

| DIRECT | | | | | |
|---|---|---|---|---|---|
| **Suppliers to Hypers** | JU | MI | RE | SN | YO |
| GO-AN-LAC | | | | | |
| GO-BINH-DUONG | | | | | |
| GO-DONG-NAI | | | | | |

**Explain from shipment to items** *in item*

| NON-DIRECT | | | | | |
|---|---|---|---|---|---|
| **Suppliers to WH** | JU | MI | RE | SN | YO |
| South WH | | | | | |

| **WH to Hypers** | South WH |
|---|---|
| GO-AN-LAC | |
| GO-BINH-DUONG | |
| GO-DONG-NAI | |

| DIRECT | | | | | |
|---|---|---|---|---|---|
| **Suppliers to Hypers** | JU | MI | RE | SN | YO |
| GO-AN-LAC | | | | | |
| GO-BINH-DUONG | | | | | |
| GO-DONG-NAI | | | | | |

## Logistics Cost Variables

- Blue Cells: That are integer variables in terms of Direct method and continuous variables in terms of Non-Direct method.
- Purple Cell: That is binary when warehouse open, we must pay additional costs.

## Transformational shipment to items

As setting constraints, we must transform into one measurement. That for demand's constraint by multiple Logistics Cost Variables with 300.

# PRESCRIPTIVE

**SOLVER**

**Objective Function**

| | | |
|---|---|---|
| MIN(TC) | | *VND 1.000* |

**subject to:**

**1. Demand Constraint**

Each cell below must be "xij" and greater than Dj

| **Total of products to transport** | JU | MI | RE | SN | YO |
|---|---|---|---|---|---|
| GO-AN-LAC | | | | | |
| GO-BINH-DUONG | | | | | |
| GO-DONG-NAI | | | | | |

**2. Mathematics**

All shipments must be integer and greater than 0

**where:**

xij, where i is supplier, j is hyper

**SOLVER**

**Objective Function**

| | | |
|---|---|---|
| MIN(TC) | | *VND 1.000* |

**subject to:**

**1. Demand Constraint**

Each cell below must be "xij" and greater than Dj

| **Total of products to transport** | JU | MI | RE | SN | YO |
|---|---|---|---|---|---|
| GO-AN-LAC | | | | | |
| GO-BINH-DUONG | | | | | |
| GO-DONG-NAI | | | | | |

**2. Product flow Constraint**

This cell below must be ∑xiw - ∑xwj = 0

| | inbound - outbound | must be |
|---|---|---|
| WH | | 0 |

**3. Warehouse Constraint**

This cell below must be less than capacity, and be "∑xiw"

| | inbound | capacity |
|---|---|---|
| Capacity | | yw*5000 |

**4. Mathematics**

All shipments would be greater than 0

Open/not Warehouse is binary {0,1}

**where:**

| | | |
|---|---|---|
| xwj, where w is warehouse, j is hyper | non-direct | outbound |
| xiw, where i is supplier, w is warehouse | non-direct | inbound |

## Direct Logistics

We must find the minimum Total Logistics Costs (TC), it be:

**MIN(TC)** = ∑ xij*DCij

Where: DCij is direct logistics costs from supply i to hyper j

## Non-direct Logistics

We must find the minimum Total Logistics Costs (TC), it be:

**MIN(TC)** = ∑ xiw*IBCiw + ∑ xwj*OBCwi + yw*8000

Where: IBCiw is inbound logistics costs from supplier i to WH

OBCwj is outbound logistics costs from WH to hyper j

# PRESCRIPTIVE

**VARIABLES'S LOGISTICS COST** *in shipment*

| DIRECT | | | | | |
|---|---|---|---|---|---|
| **Suppliers to Hypers** | JU | MI | RE | SN | YO |
| GO-AN-LAC | 5 | 30 | 25 | 30 | 60 |
| GO-BINH-DUONG | 5 | 30 | 25 | 29 | 60 |
| GO-DONG-NAI | 5 | 30 | 25 | 29 | 59 |

**SOLVER**

**Objective Function**

| | | |
|---|---|---|
| MIN(TC) | 31,247.00 | *VND 1.000* |

**subject to:**

**1. Demand Constraint**

Each cell below must be "xij" and greater than Dj

| **Total of products to transport** | JU | MI | RE | SN | YO |
|---|---|---|---|---|---|
| GO-AN-LAC | 1,500 | 9,000 | 7,500 | 9,000 | 18,000 |
| GO-BINH-DUONG | 1,500 | 9,000 | 7,500 | 8,700 | 18,000 |
| GO-DONG-NAI | 1,500 | 9,000 | 7,500 | 8,700 | 17,700 |

**2. Mathematics**

All shipments must be integer and greater than 0

**where:**

xij, where i is supplier, j is hyper

**VARIABLES'S LOGISTICS COST** *in shipment*

| NON-DIRECT | | | | | |
|---|---|---|---|---|---|
| **Suppliers to WH** | JU | MI | RE | SN | YO |
| South WH | 13 | 88 | 75 | 87 | 178 |

| **WH to Hypers** | South WH | | | | binary |
|---|---|---|---|---|---|
| GO-AN-LAC | 148 | | *about WH* | open/not | 1 |
| GO-BINH-DUONG | 147 | | additional costs | 8000 | |
| GO-DONG-NAI | 147 | | capacity | 5000 | cartons |

| NON-DIRECT | | | | | |
|---|---|---|---|---|---|
| **Suppliers to Hypers** | JU | MI | RE | SN | YO |
| GO-AN-LAC | 4.345833333 | 29.18555556 | 24.89694444 | 29.06027778 | 59.73277778 |
| GO-BINH-DUONG | 4.183055556 | 29.12444444 | 24.87916667 | 28.68833333 | 59.13027778 |
| GO-DONG-NAI | 4.035833333 | 29.57694444 | 24.86 | 28.76305556 | 58.92666667 |

**SOLVER**

**Objective Function**

| | | |
|---|---|---|
| MIN(TC) | 28,705.00 | *VND 1.000* |

**subject to:**

**1. Demand Constraint**

Each cell below must be "xij" and greater than Dj

| **Total of products to transport** | JU | MI | RE | SN | YO |
|---|---|---|---|---|---|
| GO-AN-LAC | 1,320 | 8,760 | 7,470 | 8,730 | 17,940 |
| GO-BINH-DUONG | 1,260 | 8,760 | 7,470 | 8,610 | 17,760 |
| GO-DONG-NAI | 1,230 | 8,880 | 7,470 | 8,640 | 17,700 |

**2. Product flow Constraint**

This cell below must be $\sum xiw - \sum xwj = 0$

| | inbound - outbound | must be |
|---|---|---|
| WH | 0 | 0 |

**3. Warehouse Constraint**

This cell below must be less than capacity, and be "$\sum xiw$"

| | inbound | capacity |
|---|---|---|
| Capacity | 4,410 | 5000 |

**4. Mathematics**

All shipments would be greater than 0

Open/not Warehouse is binary (0,1)

**where:**

| | | |
|---|---|---|
| xwj, where w is warehouse, j is hyper | non-direct | outbound |
| xiw, where i is supplier, w is warehouse | non-direct | inbound |

**Result of Direct Logistics**

**Result of Non-direct Logistics**

As a result, the optimize of Direct Logistics is **VND 31.247 million** and Non-direct Logistics is **VND 28.705 million** monthly.

Because the Non-direct Logistics is more optimized than Direct Logistics **9%**.

We decide **Non-direct Logistics** is the method for logistics planning for Private Label in HCMC.